

TRENTOS-M on Xilinx Zynq UltraScale+ MPSoC

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- [Avnet Ultra96](#)
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This page shall be used to provide an overview of all workpackages that are necessary for the porting of TRENTOS-M to the Xilinx Zynq UltraScale+. The reference value for stories should be that they can be achieved within two to three days.

All work related epics are grouped under the initiative

DPB-45 - Getting issue details...

STATUS

 and all stories under the epic e.g.

SEOS-2161 - Getting issue details...

STATUS

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Xilinx Zynq UltraScale+ MPSoC

Types

Information about the different MPSoC types as well as the performance of the respective FPGA part can be found in the [product selection guide](#).

- CG: 2x Cortex-A53
- EG: 4x Cortex-A53 + GPU
- EV: 4x Cortex-A53 + GPU + VCU

Hardware

- 2-4x Cortex-A53
- 2x Cortex-R5
- FPGA (with varying size depending on MPSoC type)
- Optional: GPU
- Optional: VCU
- Secure Boot
- Xilinx Memory Protection Unit (XMPU)
- Xilinx Peripheral Protection Unit (XPPU)

Misc

random notes collection

- for hardware see also [Xilinx UltraScale](#)

Product Comparison

Price Ranges

The price ranges are based either on the prices of

- a single SBC
- a SOM in combination with a Carrier Board

MPSoC	Price
XCZU2CG	from 351 €
XCZU2EG	from 468 €
XCZU3CG	from 491 €
XCZU3EG	from 224 €
XCZU4CG	from 651 €
XCZU4EG	from 1.104 €
XCZU4EV	from 585 €
XCZU5CG	from 804 €
XCZU5EG	-
XCZU5EV	from 735 €
XCZU6CG	from 999€
XCZU6EG	from 1.296 €
XCZU7CG	from 1.734 €
XCZU7EG	-
XCZU7EV	from 1.311 €
XCZU9CG	-
XCZU9EG	from 1.334 €
XCZU11EG	from 2.972 €
XCZU15EG	from 1.670 €
XCZU17EG	from 2.312 €
XCZU19EG	from 3.523 €

Single-Board-Computer (SBC)

Vendor	SBC	MPSoC	CPU	RAM (PS /PL)	QSPI Flash	Ethernet	UART	SPI	I2C	SD /MMC	USB	Debug	PMOD	FMC	Source	Release Status
Avnet	Ultra96-V2	XCZU3EG		2 GiB		-	1	via expansion	via expansion	1x microSD	3	1x JTAG via expansion	-	-	http://zedboard.org/product/ultra96-v2-development-board Expansion: • USB-UART /JTAG: AES-ACC-U96-JTAG • Mezzanine: AES-ACC-U96-ME-SK • Trend PMOD adapter Ethernet: • 1 port: SCH EDA USB-ETH V1 • 4 ports: 96B Quad Ethernet Mezzanine	
Avnet	UltraZed-EG Starter Kit	XCZU3EG-SFVA625		2 GiB	64 MiB	1x 10/100 /1000	1x USB UART			1x microSD 1x eMMC (8 GiB)		1x USB JTAG (via module)	13	-	http://microzed.org/product/ultrazed-eg-starter-kit	
Avnet	UltraZed-EV System-on-Module Starter Kit	XCZU7EV-1FBVB900													http://zedboard.org/product/ultrazed-ev-starter-kit	
Digilent	Genesys ZU-3EG	XZU3EG		4 GiB	256 MiB	1x 10/100 /1000	1x USB UART			1x microSD	2	1x JTAG 1x USB JTAG	4	1x LPC	https://store.digilentinc.com/genesys-zu-zynq-ultrascale-mpsoc-development-board/	
Digilent	Genesys ZU -5EV	XCZU5EV		4 GiB	256 MiB	1x 10/100 /1000					2		4	1x HPC 1x LPC	https://reference.digilentinc.com/reference/programmable-logic/genesys-zu/reference-manual	
HiTech Global	HTG-Z920	ZU11EG /ZU17EG /ZU19EG		2-4 GiB / up to 16 GiB		-	2x USB UART			1x microSD		1x JTAG 1x ARM Debug			http://www.hitechglobal.com/Boards/MPSOC_UltraScale+.htm	
HiTech Global	HTG-Z922	ZU11EG /ZU19EG		4 GiB / 4 GiB (up to 16 GiB)		-	2x USB UART			1x microSD		1x JTAG 1x ARM Debug			http://www.hitechglobal.com/Boards/Zynq_UltraScale+_PCIe.htm	

HiTech Global	HTG-Z999	ZU6EG /ZU9EG /ZU15EG		4 GiB / 4 GiB	-						1	1x JTAG 1x ARM Debug		1	http://www.hitechglobal.com/Boards/Zynq_ultrascale+_FMC.htm	
iWave Systems	IW-G30D-C4CG-4E002G-E008G-LCA	XCZU4CG													https://www.iwavesystems.com/product/zu7-zu5-zu4-zynq-ultrascale-mpsocsom/	4 weeks delivery
iWave Systems	IW-RainboW-G36S	XCZU4EV		2 GiB / 1 GiB	-	2x 10/100 /1000	1x Debug 1x via B2B connector	1x via B2B connect or	1x via B2B connect or	1x microSD 1x eMMC (8 GiB)	2	1x JTAG	-	-	https://www.iwavesystems.com/product/zu5-zu4-zu3-zu2-zynq-ultrascale-mpsoc-sbc/	
iWave Systems	IW-G30D-C4EV-4E002G-E008G-LCA	XCZU4EV													https://www.iwavesystems.com/product/zu7-zu5-zu4-zynq-ultrascale-mpsocsom/	4 weeks delivery
iWave Systems	IW-RainboW-G36S	XCZU5EV		2 GiB / 1 GiB	-	2x 10/100 /1000	1x Debug 1x via B2B connector	1x via B2B connect or	1x via B2B connect or	1x microSD 1x eMMC (8 GiB)	2	1x JTAG	-	-	https://www.iwavesystems.com/product/zu5-zu4-zu3-zu2-zynq-ultrascale-mpsoc-sbc/	
iWave Systems	IW-RainboW-G30D (IW-G30D-C5EV-4E004G-E008G-LCA)	XCZU5EV													https://www.iwavesystems.com/product/zu7-zu5-zu4-zynq-ultrascale-mpsocsom/	4 weeks delivery
iWave Systems	IW-RainboW-G30D (IW-G30D-C7CG-4E004G-E008G-LCA)	XCZU7CG													https://www.iwavesystems.com/product/zu7-zu5-zu4-zynq-ultrascale-mpsocsom/	4 weeks delivery
iWave Systems	IW-RainboW-G30D (IW-G30D-C7EV-4E004G-E008G-LCA)	XCZU7EV													https://www.iwavesystems.com/product/zu7-zu5-zu4-zynq-ultrascale-mpsocsom/	4 weeks delivery
iWave Systems	IW-RainboW-G35D (IW-G35D-11EG-4E004G-E008G-LCB)	XCZU11EG													https://www.iwavesystems.com/product/zu19-zu17-zu11-zynq-ultrascale-mpsocsom/	6 weeks delivery
iWave Systems	IW-RainboW-G35D (IW-G35D-19EG-4E004G-E008G-LCD)	XCZU19EG													https://www.iwavesystems.com/product/zu19-zu17-zu11-zynq-ultrascale-mpsocsom/	
iWave Systems	IW-RainboW-G35D (IW-G35D-19EG-4E004G-E008G-LCB)	XCZU19EG													https://www.iwavesystems.com/product/zu19-zu17-zu11-zynq-ultrascale-mpsocsom/	44 weeks delivery
MYIR Tech Limited	FZ3 Card - Deep Learning Accelerator Card	XCZU3EG-1SFVC784E		4 GiB	32 MiB	1x 10/100 /1000	1x USB UART			1x microSD 1x eMMC (8 GiB)	2	1x JTAG	-	-	http://www.myirtech.com/list.asp?id=630	

MYIR Tech Limited	MYD-CZU3EG Development Board	XCZU3EG-1SFVC784E		4 GiB	128 MiB	1x 10/100 /1000	1x USB UART			1x SD 1x eMMC (4 GiB)			2	1x LPC	http://www.myirtech.com/list.asp?id=613	
MYIR Tech Limited	VECP Starter Kit	XCZU3EG-1SFVC784E		4 GiB	128 MiB	2x 10/100 /1000 (one for PL)	1x USB UART			1x microSD 1x eMMC (4 GiB)	1				http://www.myirtech.com/list.asp?id=631	
MYIR Tech Limited	MYD-CZU4EV Development Board	XCZU4EV-1SFVC784E		4 GiB	128 MiB	1x 10/100 /1000	1x USB UART			1x SD 1x eMMC (4 GiB)	1	1x JTAG	2	1x LPC	http://www.myirtech.com/list.asp?id=613	
MYIR Tech Limited	FZ5 Card – AI Accelerator Card	XCZU5EV-2SFVC784E		4/8 GiB	64 MiB	1x 10/100 /1000	1x USB UART			1x microSD 1x eMMC (32 GiB)	4		-	-	http://www.myirtech.com/list.asp?id=638	
MYIR Tech Limited	FZ5 EdgeBoard AI Box	XCZU5EV-1SFVC784E		4/8 GiB	64 MiB	1x 10/100 /1000	1x USB UART			1x microSD 1x eMMC (32 GiB)	4		-	-	http://www.myirtech.com/list.asp?id=639	
Trenz	TE0802	XCZU2CG-1SBVA484E		1 GiB	32 MiB	1x 10/100 /1000	1x USB UART			1x microSD	1	1x USB JTAG	2	-	https://wiki.trenz-electronic.de/display/ID/TE0802+Resources	
Trenz	TE0803-03-3BE11-AS Starter Kit	XCZU3EG-1SFVC784E		2 GiB	128 MiB	1x 10/100 /1000				1x microSD 1x eMMC (4 GiB)	3	1x JTAG	3	1x HPC	https://wiki.trenz-electronic.de/display/ID/Starter+Kit+803	38 weeks delivery
Trenz	TE0807-03-7DE21-AS Starter Kit	XCZU7EV-1FBVB900E		4 GiB	128 MiB	1x 10/100 /1000				1x microSD 1x eMMC (4 GiB)	3	1x JTAG	3	1x HPC	https://wiki.trenz-electronic.de/display/ID/Starter+Kit+807	39 weeks delivery
Trenz	TE0808-04-9BE21-AS Starter Kit TE0808-05-9BE21-AS Starter Kit	XCZU9EG-1FFVC900E		4 GiB	128 MiB	1x 10/100 /1000				1x microSD 1x eMMC (4 GiB)	3	1x JTAG	3	1x HPC	https://wiki.trenz-electronic.de/display/ID/Starter+Kit+808	38 weeks delivery
Trenz	TE0808-04-9GI21-AS Starter Kit TE0808-05-9GI21-AS Starter Kit	XCZU9EG-2FFVC900E		4 GiB	128 MiB	1x 10/100 /1000				1x microSD 1x eMMC (4 GiB)	3	1x JTAG	3	1x HPC	https://wiki.trenz-electronic.de/display/ID/Starter+Kit+808	38 weeks delivery
Trenz	TEB0911	XCZU9EG-1FFVB1156E		8 GiB	64 MiB	1x 10/100 /1000	1x USB UART			1x SD 1x eMMC (4 GiB)	2	2x USB JTAG	-	6	https://wiki.trenz-electronic.de/display/ID/TEB0911+Resources	38 weeks delivery
Trenz	TE0808-05-BBE21-AS Starter Kit	XCZU15EG-1FFVC900E		4 GiB	128 MiB	1x 10/100 /1000				1x microSD 1x eMMC (4 GiB)	3	1x JTAG	3	1x HPC	https://wiki.trenz-electronic.de/display/ID/Starter+Kit+808	38 weeks delivery
Xilinx	ZCU104	XCZU7EV-2FFVC1156		2 GiB		1x 10/100 /1000	1x USB UART	-	-	1x microSD	1	1x USB JTAG	3	1x LPC	https://www.xilinx.com/products/boards-and-kits/zcu104.html	
Xilinx	ZCU106	XCZU7EV-2FFVC1156		4 GiB	128 MiB	1x 10/100 /1000	1x USB UART	-	via PMOD	1x SD	1	1x JTAG 1x USB JTAG 1x ARM Trace	2	2x HPC	https://www.xilinx.com/products/boards-and-kits/zcu106.html	

Xilinx	ZCU102	XCZU9EG-2FFVB1156		4 GiB / 512 MiB	128 MiB	1x 10/100 /1000	1x USB UART	-	via PMOD	1x SD	1	1x JTAG 1x USB JTAG 1x ARM Trace	2	2x HPC	https://www.xilinx.com/products/boards-and-kits/ek-u1-zcu102-g.html	
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System-on-Module (SOM)

Vendor	SOM	MPSoC	CPU	RAM (PS /PL)	QSPI Flash	Ethernet	UART	SPI	I2C	SD /MMC	USB	Debug	PMOD	FMC	Source	Release Status
Avnet	UltraZed-EG SOM	AES-ZU3EG-1-SOM-G AES-ZU3EG-1-SOM-I-G		2 GiB	64 MiB	PHY					ULPI PHY				http://microzed.org/product/ultrazed-EG	
Avnet	UltraZed-EV SOM	AES-ZU7EV-1-SOM-G AES-ZU7EV-1-SOM-I-G		4 GiB / 1 GiB	64 MiB	PHY				1x eMMC (8 GiB)	PHY				http://zedboard.org/product/ultrazed-ev	
Enclustra	Mars XU3	XCZU2CG XCZU2EG XCZU3EG		up to 4 GiB	64 MiB	PHY				1x eMMC (16 GiB)	PHY				https://www.enclustra.com/en/products/system-on-chip-modules/mars-xu3/	
Enclustra	Mercury+ XU1	XCZU6CG XCZU6EG XCZU9EG XCZU15EG		up to 8 GiB	64 MiB	2x PHY				1x eMMC (16 GiB)	4x PHY				https://www.enclustra.com/en/products/system-on-chip-modules/mercury-xu1/	
Enclustra	Mercury XU5	XCZU2CG XCZU2EG XCZU3EG XCZU4CG XCZU4EV XCZU5EV		up to 8 GiB / up to 2 GiB	64 MiB	2x PHY				1x eMMC (16 GiB)	4x PHY				https://www.enclustra.com/en/products/system-on-chip-modules/mercury-xu5/	
Enclustra	Mercury+ XU6	XCZU2CG XCZU2EG XCZU3EG XCZU4CG XCZU4EV XCZU5EV		up to 8 GiB	64 MiB	PHY				1x eMMC (16 GiB)	2x PHY				https://www.enclustra.com/en/products/system-on-chip-modules/mercury-xu6/	in development
Enclustra	Mercury+ XU7	XCZU6EG XCZU9EG XCZU15EG		up to 8 GiB / up to 4 GiB	64 MiB	2x PHY				1x eMMC (16 GiB)	4x PHY				https://www.enclustra.com/en/products/system-on-chip-modules/mercury-xu7/	
Enclustra	Mercury+ XU8	XCZU4CG XCZU5EV XCZU7EV		up to 8 GiB / up to 4 GiB	64 MiB	2x PHY				1x eMMC (16 GiB)	4x PHY				https://www.enclustra.com/en/products/system-on-chip-modules/mercury-xu8/	
Enclustra	Mercury+ XU9	XCZU4CG XCZU5EV XCZU7EV		up to 8 GiB / up to 8 GiB	64 MiB	2x PHY				1x eMMC (16 GiB)	4x PHY				https://www.enclustra.com/en/products/system-on-chip-modules/mercury-xu9/	

iVeia	Atlas-I-Z8	ZU2CG, ZU2EG, ZU3EG, ZU4EV, or ZU5EV		4 GiB / 2 GiB	128 MiB				1x microSD (optional) 1x eMMC (4 to 32 GiB)					https://iveia.com/atlas-i-z8	
iVeia	Atlas-II-Z8	ZU4CG, ZU4EV, ZU6EG, ZU7EV, ZU9EG, or ZU15EG		4 GiB / 2 GiB	128 MiB				1x microSD (optional) 1x eMMC (up to 32 GiB)					https://iveia.com/atlas-ii-z8	
iVeia	Atlas-III-Z8	ZU7EV, ZU11EG, or ZU19EG		8 GiB / 8GiB	256 MiB				1x eMMC (up to 32 GiB)					https://iveia.com/atlas-iii-z8x	
iWave Systems	iW-RainboW-G30M (iW-G30M-C4CG-4E002G-E008G-BEA)	XCZU4CG												https://www.iwavesystems.com/product/zu7-zu5-zu4-zynq-ultrascale-mpsocsom/	24 weeks delivery
iWave Systems	iW-RainboW-G30M (iW-G30M-C4CG-4E002G-E008G-BIA)	XCZU4CG													4 weeks delivery
iWave Systems	iW-RainboW-G30M (iW-G30M-C4CG-4E002G-E008G-LIA)	XCZU4CG													4 weeks delivery
iWave Systems	iW-RainboW-G30M (iW-G30M-C4EG-4E002G-E008G-BEA)	XCZU4EG													24 weeks delivery
iWave Systems	iW-RainboW-G30M (iW-G30M-C4EG-4E002G-E008G-BIA)	XCZU4EG													24 weeks delivery
iWave Systems	iW-RainboW-G30M (iW-G30M-C4EV-4E002G-E008G-BEA)	XCZU4EV													24 weeks delivery
iWave Systems	iW-RainboW-G30M (iW-G30M-C4EV-4E002G-E008G-BIA)	XCZU4EV													4 weeks delivery
iWave Systems	iW-RainboW-G30M (iW-G30M-C4EV-4E002G-E008G-LIA)	XCZU4EV													4 weeks delivery
iWave Systems	iW-RainboW-G30M (iW-G30M-C5EV-4E004G-E008G-BEA)	XCZU5EV													24 weeks delivery

iWave Systems	iW- RainboW -G30M (iW- G30M- C5EV- 4E004G- E008G- BIA)	XCZU5EV														4 weeks delivery
iWave Systems	iW- RainboW -G30M (iW- G30M- C5EV- 4E004G- E008G- LIA)	XCZU5EV														4 weeks delivery
iWave Systems	iW- RainboW -G30M (iW- G30M- C7CG- 4E004G- E008G- BEA)	XCZU7CG														24 weeks delivery
iWave Systems	iW- RainboW -G30M (iW- G30M- C7CG- 4E004G- E008G- BIA)	XCZU7CG														4 weeks delivery
iWave Systems	iW- RainboW -G30M (iW- G30M- C7CG- 4E004G- E008G- LIA)	XCZU7CG														4 weeks delivery
iWave Systems	iW- RainboW -G30M (iW- G30M- C7EV- 4E004G- E008G- BEA)	XCZU7EV														4 weeks delivery
iWave Systems	iW- RainboW -G30M (iW- G30M- C7EV- 4E004G- E008G- LEA)	XCZU7EV														4 weeks delivery
iWave Systems	iW- RainboW -G30M (iW- G30M- C7EV- 4E004G- E008G- BIA)	XCZU7EV														24 weeks delivery
iWave Systems	iW- RainboW -G35M (iW- G35M- 11EG- 4E004G- E008G- LIA)	XCZU11EG														4 weeks delivery
iWave Systems	iW- RainboW -G35M (iW- G35M- 11EG- 4E004G- E008G- BIA)	XCZU11EG														4 weeks delivery
iWave Systems	iW- RainboW -G35M (iW- G35M- 17EG- 4E004G- E008G- BIA)	XCZU17EG														4 weeks delivery

iWave Systems	iW-RainboW-G35M (iW-G35M-19EG-4E004G-E008G-BIA)	XCZU19EG													4 weeks delivery
iWave Systems	iW-RainboW-G35M (iW-G35M-19EG-4E004G-E008G-LEA)	XCZU19EG													7 weeks delivery
Knowledge Resources	KRM-3ZU2CG	XCZU2CG		2 GiB / 512 MiB											
Knowledge Resources	KRM-3ZU3CG	XCZU3CG		2 GiB / 512 MiB											
Knowledge Resources	KRM-3ZU4CG	XCZU4CG		2 GiB / 512 MiB											
Knowledge Resources	KRM-3ZU5CG	XCZU5CG		2 GiB / 512 MiB											
Knowledge Resources	KRM-5ZU9EG	XCZU9EG		8 GiB / 8 GiB		5x PHY (PS) 4x PHY (PL)									
MYIR Tech Limited	MYC-CZU3EG CPU Module	XCZU3EG-1SFVC784E		4 GiB	128 MiB	PHY			1x eMMC (4 GiB)	PHY				http://www.myirtech.com/list.asp?id=612	
MYIR Tech Limited	MYC-CZU4EV CPU Module	XCZU4EV-1SFVC784I		4 GiB	128 MiB	PHY			1x eMMC (4 GiB)	PHY				http://www.myirtech.com/list.asp?id=612	
Reflex CES	Zeus Zynq UltraScale+ module	XCZU11EG		8 GiB / 8 GiB	256 MiB	PHY			1x eMMC (128 GiB)					https://www.reflexces.com/modules/xilinx-zynq-ultrascale-mpsoc/zeus-zynq-ultrascale-mpsoc-system-on-module	
SoC-e	SMARTmpsoc module	XCZU3EG		2 GiB	128 MiB				1x eMMC (16 GiB)	PHY				https://soc-e.com/products/smartmpsoc-module-zynq-ultrascale-som-for-time-aware-networking/	
Topic Embedded Systems	Miami MPSoC	ZU6/ZU9 /ZU15		2 GiB	64 MiB									https://topic.nl/en/products/system-on-modules/miami-mpsoc	
Topic Embedded Systems	Miami MPSoC Plus	ZU6/ZU9 /ZU15		4 GiB (up to 8 / 16 GiB)	64 MiB									https://topic.nl/en/products/system-on-modules/miami-mpsoc-plus	
Trenz	TE0820-04-2AE21FA (4 x 5)	XCZU2CG-1SFVC784E		2 GiB	128 MiB				1x eMMC (8 GiB)						from 04.06.2021
Trenz	TE0820-04-2AI21FA (4 x 5)	XCZU2CG-1SFVC784I		2 GiB	128 MiB				1x eMMC (8 GiB)						from 08.02.2021
Trenz	TE0821-01-2AE31KA (4 x 5)	XCZU2CG-1SFVC784E		4 GiB	128 MiB				1x eMMC (64 GiB)						
Trenz	TE0820-04-2BE21FA (4 x 5)	XCZU2EG-1SFVC784E		2 GiB	128 MiB				1x eMMC (8 GiB)						from 29.04.2021
Trenz	TE0820-04-2BE21FL (4 x 5)	XCZU2EG-1SFVC784E		2 GiB	128 MiB				1x eMMC (8 GiB)						38 weeks delivery
Trenz	TE0820-04-2BI21FL (4 x 5)	XCZU2EG-1SFVC784I		2 GiB	128 MiB				1x eMMC (8 GiB)						38 weeks delivery

Trenz	TE0820-04-2BI21FA (4 x 5)	XCZU2EG-1SFVC784I		2 GiB	128 MiB					1x eMMC (8 GiB)						38 weeks delivery
Trenz	TE0820-04-3AE21FA (4 x 5)	XCZU3CG-1SFVC784E		2 GiB	128 MiB					1x eMMC (8 GiB)						from 18.02.2021
Trenz	TE0821-01-3AE31KA (4 x 5)	XCZU3CG-1SFVC784E		4 GiB	128 MiB					1x eMMC (64 GiB)						from 29.01.2021
Trenz	TE0823-01-3PIU1FL (4 x 5)	XCZU3CG-L1SFVC784I		1 GiB	128 MiB					1x eMMC (8 GiB)						38 weeks delivery
Trenz	TE0823-01-3PIU1FA (4 x 5)	XCZU3CG-L1SFVC784I		1 GiB	128 MiB					1x eMMC (8 GiB)						from 16.07.2021
Trenz	TE0821-01-3BE21FL (4 x 5)	XCZU3EG-1SFVC784E		2 GiB	128 MiB					1x eMMC (8 GiB)						from 08.07.2021
Trenz	TE0820-04-3BE21FA (4 x 5)	XCZU3EG-1SFVC784E		2 GiB	128 MiB					1x eMMC (8 GiB)						from 18.02.2021
Trenz	TE0820-04-3BE21FL (4 x 5)	XCZU3EG-1SFVC784E		2 GiB	128 MiB					1x eMMC (8 GiB)						from 18.02.2021
Trenz	TE0821-01-3BE21FA (4 x 5)	XCZU3EG-1SFVC784E		2 GiB	128 MiB					1x eMMC (8 GiB)						from 18.02.2021
Trenz	TE0821-01-3BI21FA (4 x 5)	XCZU3EG-1SFVC784I		2 GiB	128 MiB					1x eMMC (8 GiB)						from 26.02.2021
Trenz	TE0820-04-4AE21FA (4 x 5)	XCZU4CG-1SFVC784E		2 GiB	128 MiB					1x eMMC (8 GiB)						from 12.03.2021
Trenz	TE0820-03-4DE21FA (4 x 5)	XCZU4EV-1SFVC784E		2 GiB	128 MiB					1x eMMC (8 GiB)						
Trenz	TE0820-04-4DE21FA (4 x 5)	XCZU4EV-1SFVC784E		2 GiB	128 MiB					1x eMMC (8 GiB)						
Trenz	TE0820-04-4DI21FA (4 x 5)	XCZU4EV-1SFVC784I		2 GiB	128 MiB					1x eMMC (8 GiB)						
Trenz	TE0820-04-5DI21FA (4 x 5)	XCZU5EV-1SFVC784I		2 GiB	128 MiB					1x eMMC (8 GiB)						from 09.04.2021
Trenz	TE0803-03-2AE11-A (5,2 x 7,6)	XCZU2CG-1SFVC784E		2 GiB	128 MiB											
Trenz	TE0803-03-2BE11-A (5,2 x 7,6)	XCZU2EG-1SFVC784E		2 GiB	128 MiB											
Trenz	TE0803-03-3AE11-A (5,2 x 7,6)	XCZU3CG-1SFVC784E		2 GiB	128 MiB											
Trenz	TE0803-03-3BE11-A (5,2 x 7,6)	XCZU3EG-1SFVC784E		2 GiB	128 MiB											
Trenz	TE0803-03-4AE11-A (5,2 x 7,6)	XCZU4CG-1SFVC784E		2 GiB	128 MiB											from 16.04.2021
Trenz	TE0807-03-4AI21-A (5,2 x 7,6)	XCZU4CG-1FBVB900I		4 GiB	128 MiB											
Trenz	TE0803-03-4BE11-A (5,2 x 7,6)	XCZU4EG-1SFVC784E		2 GiB	128 MiB											from 11.06.2021

Trenz	TE0803-03-4GE21-L (5,2 x 7,6)	XCZU4EG-2SFVC784E		4 GiB	128 MiB											
Trenz	TE0807-03-4BE21-A (5,2 x 7,6)	XCZU4EG-1FBVB900E		4 GiB	128 MiB											31 weeks delivery
Trenz	TE0803-03-4DE11-A (5,2 x 7,6)	XCZU4EV-1SFVC784E		2 GiB	128 MiB											from 30.04.2021
Trenz	TE0803-03-4DE21-L (5,2 x 7,6)	XCZU4EV-1SFVC784E		4 GiB	128 MiB											from 30.04.2021
Trenz	TE0803-03-5DE11-A (5,2 x 7,6)	XCZU5EV-1SFVC784E		2 GiB	128 MiB											from 26.02.2021
Trenz	TE0808-04-6BE21-A (5,2 x 7,6)	XCZU6EG-1FFVC900E		4 GiB	128 MiB											27 weeks delivery
Trenz	TE0808-05-6BE21-A (5,2 x 7,6)	XCZU6EG-1FFVC900E		4 GiB	128 MiB											from 09.04.2021
Trenz	TE0808-04-6BE21-L (5,2 x 7,6)	XCZU6EG-1FFVC900E		4 GiB	128 MiB											from 04.06.2021
Trenz	TE0808-05-6BE21-L (5,2 x 7,6)	XCZU6EG-1FFVC900E		4 GiB	128 MiB											from 04.06.2021
Trenz	TE0807-03-7AI21-A (5,2 x 7,6)	XCZU7CG-1FBVB900I		4 GiB	128 MiB											from 08.02.2021
Trenz	TE0807-03-7DE21-A (5,2 x 7,6)	XCZU7EV-1FBVB900E		4 GiB	128 MiB											
Trenz	TE0807-03-7DI21-A (5,2 x 7,6)	XCZU7EV-1FBVB900I		4 GiB	128 MiB											from 08.02.2021
Trenz	TE0808-04-9BE21-A (5,2 x 7,6)	XCZU9EG-1FFVC900E		4 GiB	128 MiB											from 29.01.2021
Trenz	TE0808-05-9BE21-A (5,2 x 7,6)	XCZU9EG-1FFVC900E		4 GiB	128 MiB											from 12.02.2021
Trenz	TE0808-04-9BE21-L (5,2 x 7,6)	XCZU9EG-1FFVC900E		4 GiB	128 MiB											from 30.04.2021
Trenz	TE0808-05-9BE21-L (5,2 x 7,6)	XCZU9EG-1FFVC900E		4 GiB	128 MiB											31 weeks delivery
Trenz	TE0808-04-9GI21-A (5,2 x 7,6)	XCZU9EG-2FFVC900I		4 GiB	128 MiB											
Trenz	TE0808-05-9GI21-A (5,2 x 7,6)	XCZU9EG-2FFVC900I		4 GiB	128 MiB											from 21.05.2021
Trenz	TE0808-04-BBE21-A (5,2 x 7,6)	XCZU15EG-1FFVC900E		4 GiB	128 MiB											from 22.01.2021
Trenz	TE0808-05-BBE21-A (5,2 x 7,6)	XCZU15EG-1FFVC900E		4 GiB	128 MiB											from 30.06.2021

Carrier Boards

Vendor	Carrier Board	SOM Type	CPU	RAM (PS /PL)	QSPI Flash	Ethernet	UART	SPI	I2C	SD /MMC	USB	Debug	PMOD	FMC	Source	Release Status
Avnet	UltraZed-EG IO Carrier Card (AES-ZU-IOCC-G)	UltraZed-EG SOM													http://zedboard.org/product/ultrazed-eg-io-carrier-card	
Avnet	UltraZed PCIe Carrier Card (AES-ZU-PCIECC-G)	UltraZed-EG SOM													http://microzed.org/product/ultrazed-eg-pcie-carrier-card	
Avnet	UltraZed-EV Carrier Card (AES-ZUEV-CC-G)	UltraZed-EV SOM													http://zedboard.org/product/ultrazed-ev-carrier-card	
Enclustra	Mars ST3	Mars SOM				1x 10/100 /1000				1x microSD					https://www.enclustra.com/en/products/baseboards/mars-st3/	
Enclustra	Mars EB1	Mars SOM				1x 10/100 /1000				1x microSD					https://www.enclustra.com/en/products/baseboards/mars-eb1/	
Enclustra	Mercury+ PE1-200 /300/400	Mercury SOM				2x 10/100 /1000	1x USB UART			1x microSD	4	1x USB JTAG		1x HPC 3x LPC	https://www.enclustra.com/en/products/baseboards/mercury-pe1-200-300-400/	
Enclustra	Mercury+ ST1	Mercury SOM				2x 10/100 /1000				1x microSD	2			1x HPC	https://www.enclustra.com/en/products/baseboards/mercury-st1/	
Knowledge Resources	KRC3701_CK	KRM-3ZUxxx				1x 10/100 /1000	2x USB UART			1x microSD	4		4	1x HPC		
Knowledge Resources	KRC_5700															coming soon
MYIR Tech Limited	MYC-CZU3EG	XCZU3EG-1SFVC78 4E		4 GiB	128 MiB					1x eMMC (4 GiB)						
Reflex CES	Zeus Zynq UltraScale+ board	Zeus Zynq UltraScale+ module				1x 10/100 /1000	1x USB UART				3			1x HPC 1x LPC	https://www.reflexces.com/modules/xilinx-zynq-ultrascale-mpsoc/zeus-zynq-ultrascale-mpsoc-som-carrier-board	
Topic Embedded Systems	Florida-GEN Florida-MED Florida PCIe	Miami SOM (only for Zynq-7000 ???)				1x 10/100 /1000	4x USB UART				1	1x JTAG	3		https://www.xilinx.com/products/boards-and-kits/1-foivpn.html	
Trenz	TEB0707	Trenz 4 x 5				1x 10/100 /1000				1x microSD	2					27 weeks delivery
Trenz	TEF1002-02	Trenz 4 x 5				1x 10/100 /1000	1x USB UART			1x microSD	1	1x USB JTAG				from 30.04.2021
Trenz	TE0706	Trenz 4 x 5				1x 10/100 /1000	1			1x microSD	1	1x JTAG				from 26.03.2021
Trenz	TE0703	Trenz 4 x 5				1x 10/100 /1000	1x USB UART			1x microSD	1	1x USB JTAG				from 12.02.2021
Trenz	TE0705	Trenz 4 x 5				1x 10/100 /1000				1x SD		1x USB JTAG				from 12.03.2021
Trenz	TE0701	Trenz 4 x 5				1x 10/100 /1000				1x SD	1	1x USB JTAG	2			from 26.02.2021

Trenz	TEBF0808	Trenz 5,2 x 7,6				1x 10/100 /1000				1x microSD 1x eMMC (4 GiB)	3	1x JTAG	3	1x HPC		from 19.03.2021
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Product Selection

SoC Selection

We will concentrate on the following MPSoC types:

MPSoC	CPU	System Logic Cells (K)	CLB Flip-Flops (K)	CLB LUTs (K)	Total Block RAM (Mb)	DSP Slices
XCZU2CG	2x Cortex-A53 (up to 1,3 GHz)	103	94	47	5,3	240
XCZU3EG	4x Cortex-A53 (up to 1,5 GHz)	154	141	71	7,6	360
XCZU9EG	4x Cortex-A53 (up to 1,5 GHz)	600	548	274	32,1	2.520

Single-Board-Computer (SBC) Selection

The sel4 already provides support for the Xilinx Zynq UltraScale+ SoC in a limited form and based on specific development boards. According to Data61 (<https://docs.sel4.systems/Hardware/>) and DornerWorks, the sel4 currently supports the following reference SBCs:

- Xilinx ZCU102
- Avnet Ultra96 (v1/v2)

So, all in all, the following two platforms have to be investigated:

Vendor	SBC	MPSoC	CPU	RAM (PS /PL)	QSPI Flash	Ethernet	UART	SPI	I2C	SD /MMC	USB	Debug	PMOD	FMC	Source	Release Status
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Avnet	Ultra96-V2	XCZU3EG		2 GB		-	1	via expansion	via expansion	1x microSD	2 ports + 1 via expansion	1x JTAG via expansion	-	-	http://zedboard.org/product/ultra96-v2-development-board Expansion: • USB-UART /JTAG: AES-ACC - U96-JTAG • Mezzanine: AES-ACC - U96-ME-SK • Trenz PMOD adapter Ethernet: • 1 port: SCH EDA USB-ETH V1 (USB bridge) • 4 ports : 96B Quad Ethernet Mezzanine (native) commercial grade vs industrial grade version: • https://www.element14.com/community/groups/fpga-blog/2020/02/20/industrial-temperature-ultra96-v2 • passive heat sink AES-ACC - U96-PHS1	
Xilinx	ZCU102	XCZU9EG-2FFVB1156		4 GiB / 512 MiB	128 MiB	1x 10/100/1000	1x USB UART	-	via PMOD	1x SD	1	1x JTAG 1x USB JTAG 1x ARM Trace	2	2x HPC	https://www.xilinx.com/products/boards-and-kits/ek-u1-zcu102-g.html	

Availability at HC

Board	Amount (Total)	Amount (Available)	Location
Xilinx ZCU102	7	3	Development Boards Inventory
Avnet Ultra96 (Rev v2)	4	2 with JTAG adapter and 4 port ethernet addon board https://opsero.com/product/96b-quad-ethernet-mezzanine (Unknown User (axehei01)) 1 with JTAG adapter (Unknown User (micpet01)) 1 with non-working UART output (Axel Heider)	

Single-Board-Computer (SBC) or Carrier Board + SOM Alternatives

Besides the SBCs mentioned above, the following alternatives might be evaluated for purchase as well:

Vendor	SBC	MPSoC	RAM (PS /PL)	QSPI Flash	Ethernet	UART	SPI	I2C	SD /MMC	USB	Debug	PMOD	FMC	Source	Release Status	Compan Location
Enclustra	Mars ST3 Mars XU3	XCZU2CG	up to 4 GiB	64 MiB	1x 10/100 /1000				1x microSD 1x eMMC (16 GiB)							Switzerland
Enclustra	Mercury+ ST1 Mercury XU5	XCZU2CG	up to 8 GiB / up to 2 GiB	64 MiB	2x 10/100 /1000				1x microSD 1x eMMC (16 GiB)	2			1x HPC			Switzerland
Knowledge Resources	KRC370 1_CK KRM-3ZU2CG	XCZU2CG	2 GiB / 512 MiB		1x 10/100 /1000	2x USB UART			1x microSD	4		4	1x HPC			Switzerland
MYIR Tech Limited	FZ3 Card - Deep Learning Accelerat or Card	XCZU3EG-1SFVC784E	4 GiB	32 MiB	1x 10/100 /1000	1x USB UART			1x microSD 1x eMMC (8 GiB)	2	1x JTAG	-	-	http://www.myrirtech.com/list.asp?id=630		China
Trenz	TE0802	XCZU2CG-1SBVA484E	1 GiB	32 MiB	1x 10/100 /1000	1x USB UART			1x microSD	1	1x USB JTAG	2	-	https://wiki.trenz-electronic.de/display/ID/TE0802+Resources		Germany

QEMU support

QEMU already supports the Xilinx Zynq UltraScale+ in form of the target **xlnx-zcu102**, which is a virtual pendant of the Xilinx ZCU102 board.

Versions

- Xilinx: <https://github.com/Xilinx/qemu>
- Mainline:
 - <https://git.qemu.org/?p=qemu.git>
 - <https://github.com/qemu/qemu>

Xilinx QEMU

Detailed information on the Xilinx QEMU version and the modeled Zynq UltraScale+ devices can be found in

- <https://xilinx-wiki.atlassian.net/wiki/spaces/A/pages/821395464/QEMU+User+Documentation>
- PMU: <https://xilinx-wiki.atlassian.net/wiki/spaces/A/pages/18842214/MPSoC+Power+Management>
- the document [UG1169](#)

QEMU SystemC Co-Simulation

- Xilinx
 - Documentation
 - <https://xilinx-wiki.atlassian.net/wiki/spaces/A/pages/862421112/Co-simulation>
 - http://users.ece.utexas.edu/~gerstl/ee382m_f18/labs/QEMU_SystemC_Tutorial.htm
 - Software

- <https://github.com/Xilinx/libsystemctlm-soc>
 - <https://github.com/Xilinx/systemctlm-cosim-demo>
 - Third-party
 - Aldec: https://www.aldec.com/en/solutions/functional_verification/qemu_co_sim

Mainline QEMU

Features

- v2.4: New board model xlnx-ep108 -> later replaced by "xlnx-zcu102"
- v2.5: support for AHCI
- v2.6:
 - The "xlnx-zynqmp" machine type now includes SPI controllers
 - The "xlnx-ep108" machine type now supports SPI flash
- v.2.7:
 - New Xilinx Zynq ZCU102 board (-M xlnx-zcu102)
 - Xilinx Zynq boards have experimental support for ARM Security Extensions.
 - Xilinx Zynq MP supports DisplayPort (graphics and audio) and DDC (used for EDID info).
- v3.0: The xlnx-zynqmp machine uses a Cortex-R5F CPU rather than a Cortex-R5.
- v3.1: New virtual Xilinx Versal machine model: "xlnx-versal-virt"
- v5.0: The versal board now implements the ADMA DMA controllers
- v5.1: xlnx-versal-virt: machine now supports RTC and SD
- v6.0:
 - xlnx-zynqmp boards now support the Xilinx ZynqMP CAN controllers
 - the xlnx-versal board now has USB support

Emulated Hardware

QEMU (version ...) already supports the Xilinx Zynq UltraScale+ in form of the target **xlnx-zcu102**, which is a virtual pendant of the Xilinx ZCU102 board.

HW	Mainline QEMU		Xilinx QEMU	
	Amount	Source	Amount	Source
Board				
Clock				
Reset (SRC)				
Serial (UART)				
Timer				
Watchdog				
NIC				
Storage (SD)				
Storage (USB)				
GPIO				
SPI				
I2C				
RNGC				

Xilinx Hardware IP & Device Drivers

The following table provides an overview and comparison of the hardware IP used by Xilinx for the Zynq-7000 and the Zynq UltraScale+. The information originates from the TRMs of the two SoCs and subsequent online research (e.g. at ARM). An overview of baremetal and Linux drivers provided by Xilinx can be found here:

- Baremetal Drivers: <https://xilinx-wiki.atlassian.net/wiki/spaces/A/pages/18841745/Baremetal+Drivers+and+Libraries>
- Linux Drivers: <https://xilinx-wiki.atlassian.net/wiki/spaces/A/pages/18841873/Linux+Drivers>

HW Type	Xilinx Zynq-7000	Name in .dts	Xilinx Zynq UltraScale+	Name in .dts	Baremetal drivers	Linux drivers	Documents
UART	Cadence	xlnx, xuartps	Cadence (r113)	xlnx, xuartps	Zynq-7000 /Zynq UltraScale+	drivers/tty/serial /xilinx_uartps.c (Zynq-7000 /Zynq UltraScale+)	

AXI UART Lite	Xilinx LogiCORE™ IP AXI Universal Asynchronous Receiver Transmitter (UART) Lite interface		Xilinx LogiCORE™ IP AXI Universal Asynchronous Receiver Transmitter (UART) Lite interface		Zynq-7000 /Zynq UltraScale+	drivers/tty/serial/uartlite.c (Zynq-7000/Zynq UltraScale+)	AXI UART Lite v2.0
GPIO		xlnx, zynq-gpio-1.0		xlnx, zynqmp-gpio-1.0	Zynq-7000 /Zynq UltraScale+	drivers/gpio/gpio-zynq.c (Zynq-7000/Zynq UltraScale+)	
I2C Controllers	Cadence	cdns, i2c-r1p10	Cadence (r114_f0100_final)	cdns, i2c-r1p10 cdns, i2c-r1p14	Zynq-7000 /Zynq UltraScale+	drivers/i2c/busses/i2c-cadence.c (Zynq-7000 /Zynq UltraScale+)	
SPI Controllers	Cadence	xlnx, zynq-spi-r1p6	Cadence (r109)	cdns, spi-r1p6	Zynq-7000 /Zynq UltraScale+	drivers/spi/spi-cadence.c (Zynq-7000/Zynq UltraScale+)	
Mux		xlnx, pinctrl-zynq					
Ethernet	Cadence	cdns, zynq-gem cdns, gem	Cadence (r2p03)	cdns, zynqmp-gem cdns, gem	Zynq-7000 /Zynq UltraScale+	drivers/net/ethernet/xilinx/xilinx_emacps.c (Zynq-7000/Zynq UltraScale+) drivers/net/ethernet/cadence/macb.c (Zynq-7000/Zynq UltraScale+)	
USB	Synopsys USB 2.0 Synopsys DRD controller (former Chipidea)	xlnx, zynq-usb-2.20a chipidea, usb2	Synopsys USB 3.0 Synopsys DRD controller (2.90a)	snps, dwc3	Zynq-7000 Zynq UltraScale+	Zynq-7000 (Host) - drivers/usb/host/zynq-dr-of.c - drivers/usb/host/ehci-zynq.c - drivers/usb/chipidea/ Zynq-7000 (Device) - drivers/usb/gadget/zynq_udc.c - drivers/usb/chipidea/udc.c - drivers/usb/chipidea/ci_hdrc_usb2.c Zynq-7000 (OTG) - drivers/usb/phy/phy-ulpi.c - drivers/usb/chipidea/ Zynq UltraScale+ - drivers/usb/dwc3	
SD/SDIO /eMMC Controllers	Arasan SD 2.0 host controller	arasan, sdhci-8.9a	Arasan SD 3.0 host controller	arasan, sdhci-8.9a	Zynq-7000 Zynq UltraScale+	drivers/mmc/host/sdhci-of-arasan.c (Zynq-7000/Zynq UltraScale+)	
Triple Timer Counter (TTC)	Cadence	cdns, ttc	Cadence (r2)	cdns, ttc	Zynq-7000 /Zynq UltraScale+		
ARM Global Timer	ARM Cortex-A9 MPCore Global Timer (r3p0)	arm, cortex-a9-global-timer	ARM Generic Timer	arm, armv8-timer			
Level 2 Cache Controller	ARM AMBA Level 2 Cache Controller (L2C-310)	arm, pl310-cache					
SMMU			ARM CoreLink MMU-500 System Memory Management Unit (r2p1)	arm, mmu-500			
Interrupt Controller	ARM Generic Interrupt Controller v1.0 PrimeCell Generic Interrupt Controller (PL390)	arm, cortex-a9-gic	Cortex-A53: CoreLink GIC-400 Generic Interrupt Controller (r0p1) Cortex-R5F: PrimeCell Generic Interrupt Controller (PL390) (r0p0)	arm, gic-400			
DMA Controller	ARM PrimeCell DMA Controller (PL330)						
Memory Controller	ARM PrimeCell Static Memory Controller (PL350 series)						

Existing seL4 support

The mainline seL4 already provides support for the Xilinx Zynq UltraScale+ in a limited form and based on specific development boards.

SBC

According to <https://docs.sel4.systems/Hardware/>, seL4 currently supports the following Xilinx Zynq UltraScale+ platform:

- Xilinx ZCU102 (<https://docs.sel4.systems/Hardware/ZCU102.html>)

Based on the source code (and information available from the company DornerWorks, there also seems to be support for the low cost SBC

- Avnet Ultra96 (<https://dornerworks.com/blog/sel4-configurations/>) in Versions v1 (and v2 by DornerWorks)
- Source code containing the DornerWorks modifications available at <https://bitbucket.hensoldt-cyber.systems/projects/DW> (mainly https://bitbucket.hensoldt-cyber.systems/projects/DW/repos/sel4_hypervisor/browse?at=refs%2Fheads%2Ftrentos)

Resources

- Technical Reference Manual (TRM): https://www.xilinx.com/support/documentation/user_guides/ug1085-zynq-ultrascale-trm.pdf
- Software Developer Guide: https://www.xilinx.com/support/documentation/sw_manuals/xilinx2020_2/ug1137-zynq-ultrascale-mpsoc-swdev.pdf

Device drivers

CMake-build system

Data61	DornerWorks	Comment	Status	Task
/tools/sel4/cmake-tool /helpers /application_settings.cmake		currently only AARCH32 support for zynqmp target		check if adaptation is required for 64-bit as well
/tools/sel4/cmake-tool /helpers/simulation.cmake		currently no QEMU target available for zynqmp , only for xilinx-zynq-a9 (Zynq-7000)		provide support for the QEMU platform xlnx-zcu102 (Zynq UltraScale+), which is only available as 64-bit (via command qemu-system-aarch64)

Elfloader

Data61	DornerWorks	Comment	Status	Task
/tools/sel4/elfloader-tool /include/plat/zynqmp /sys_fputc.h		-		-
/tools/sel4/elfloader-tool /src/arch-arm/armv/armv8-a/64/		Support for zynqmp was restructured in sel4 v12 (from a separate folder to general driver); compatibility String arm,psci-0.2 (see zynqmp.dts, ultra96.dts)		-
/tools/sel4/elfloader-tool /src/arch-arm/drivers/smp-psci.c	-			
/tools/sel4/elfloader-tool /src/drivers/uart/xilinx-uart.c	-	driver checks for compatibility String xlnx,xuartps , which is available in • zynq7000.dts • zynqmp.dts • ultra96.dts		-
/tools/sel4/elfloader-tool /src/plat/	-	Support for zynqmp was restructured in sel4 v12 only a folder zynq7000 , which contains the platform_init.c ; original logic was transferred to /tools/sel4/elfloader-tool/include/plat/zynqmp/sys_fputc.h (see above)		check if an equivalent platform_init.c is required for zynqmp

seL4

Data61	DornerWorks	Comment	Status	Task
/kernel/src /arch/arm /machine/		in l2c_310.c there is a reference to CONFIG_PLAT_ZYNQ7000 , but there is no equivalent in l2c_nop.c for the Zynq UltraScale+		check if a CONFIG_PLAT_ZYNQMP is required in a separate driver for the L2 Cache Controller or if l2c_nop.c is sufficient

/kernel/src /drivers /serial /xuartps.c		driver for the Cadence UART IP		-
/kernel/src /drivers /serial /config. cmake		includes the compatibility String xlnx , xuartps for registering the driver available in xuartps.c		-
/kernel/src /drivers /smmu /config. cmake	-	includes the compatibility String arm , mmu-500 for registering the driver available in smmu_v2.c		-
/kernel/src /drivers/timer /config. cmake		includes compatibility String arm , armv8-timer for registering the driver available in generic_timer.c		-
/kernel/src /plat/zynqmp /config. cmake	- a new platform (KernelPlatformUltra96v2, PLAT_ZYNQMP_ULTRA96_V2) was added (in addition to the already available one) - usage of the new src/plat/zynqmp/overlay-ultra96v2.dts - references src/arch/arm/machine/smmu.c	includes references to - ARM Generic Interrupt Controller: src/arch/arm/machine/gic_v2.c - ARM L2 Cache Controller: src/arch/arm/machine/l2c_nop.c According to the ARM documentation , the GIC-400 implements the GICv2 architecture.		- Data61 - Question: Zynq-7000 uses GIC v1.0 and Zynq UltraScale+ uses GIC-400, but the driver used in sel4 is the same for both (gic_v2.c) - Check if the Zynq UltraScale+ L2 Cache Controller is supported adequately by l2c_nop.c - DornerWorks - Check if the support for the Avnet Ultra96 v2 is required - Compare the SMMU support provided by DornerWorks (/kernel/src/arch/arm/machine/smmu.c) with the one available from Data61 (/kernel/src/arch/arm/object/smmu.c)
/kernel/src /plat/zynqmp /overlay- zynqmp.dts		The content of overlay-zynqmp.dts differs between the Data61 and the DornerWorks version.		check if the Data61 overlay-zynqmp.dts needs adaptation
	/kernel/src/plat/zynqmp/overlay-ultra96v2.dts	DornerWorks offers a separate overlay file for the Avnet Ultra96 v2.		check if the overlay-ultra96v2.dts has to be adopted for Data61
/kernel/src /plat/zynqmp /overlay- zynqmp32. dts	-	-		-
/kernel/tools /hardware. yaml		-		check if some compatibility Strings have to be added here (see Nitrogen6_SoloX porting as a reference)
/kernel/tools /dts/ultra96. dts		.dts file for the Avnet Ultra96 SBC		compare the .dts files between the DornerWorks and Data61 versions
/kernel/tools /dts/zynqmp. dts		.dts file for the Xilinx ZCU102 SBC		compare the .dts files between the DornerWorks and Data61 versions
/kernel/tools /hardware /irq.py	-	-		check if some compatibility Strings have to be added to CONTROLLERS Note: arm, gic-400 is already existing there

Libraries

The sel4 libraries can be split into

- libplatsupport (sel4_util_libs)
- libethdrivers (sel4_util_libs)
- libsdhcd drivers (projects_libs)

libplatsupport

Data61	DornerWorks	Comment	Status	Task
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/libs/sel4_util_libs /libplatsupport /CMakeLists.txt		-		see Bitbucket branch wip-axel-ultra96 in folder sdk-sel4-camkes: • Check for KernelPlatformUltra96 and KernelPlatformZCU102 ◦ KernelPlatformUltra96 is defined in /kernel/src/plat/zynqmp/config.cmake ◦ KernelPlatformZCU102 is not existing right now is KernelPlatformZynqmp meant ? • Add selection based on this config to define which .c files have to be included during compile
/libs/sel4_util_libs /libplatsupport /mach_include/zynq/		Includes header files for drivers that are generic for the general Xilinx Zynq architecture, meaning both the Zynq-7000 and the Zynq UltraScale+ • AXI Uartlite • GPIO • I2C • Mux • Timer ◦ Triple Timer Counter (TTC) ◦ see compatibility String cdns, ttc in the zynqmp.dts and ultra96.dts		check the functionality of the drivers provided
/libs/sel4_util_libs /libplatsupport/plat_include /zynqmp/platsupport/plat/	DornerWorks code includes additional logic for the Avnet Ultra96 v2 within serial.h	Includes header files for drivers that are specific to Zynq UltraScale+ • Clock • I2C • UART ◦ UART0 (default) ◦ UART1 • Reset ◦ System Level Control Registers (slcr) • Timer Includes header files for drivers that are specific to Zynq-7000 • Clock • I2C • UART • Reset ◦ System Level Control Registers (slcr) • Timer • TMU (temperature sensor)		• check the functionality of the drivers provided • check if modifications provided by DornerWorks have to be adopted to the Data61 code • check if TMU is available at Xilinx Zynq UltraScale+
/libs/sel4_util_libs /libplatsupport/src/arch/arm/	DornerWorks code includes an additional configuration CONFIG_PLAT_ZYNQMP within delay.c			check if modifications provided by DornerWorks have to be adopted to the Data61 code
/libs/sel4_util_libs /libplatsupport/src/mach/zynq/		Includes drivers that are generic for the general Xilinx Zynq architecture, meaning both the Zynq-7000 and the Zynq UltraScale+ • AXI Uartlite • Chardev • Itimer (using the TTC) • Mux • Serial • Timer (using the TTC)		check the functionality of the drivers provided

/libs/sel4_util_libs /libplatsupport/src/plat /zynqmp/		Includes drivers that are specific to Zynq UltraScale+ • Clock		• check the functionality of the driver provided • check if TMU is available at Xilinx Zynq UltraScale+ (XADC and Devcfg not available)
/libs/sel4_util_libs /libplatsupport/src/plat /zynq7000/		Includes drivers that are specific to Zynq-7000 • Clock • Device Configuration Interface (devcfg) • Reset ◦ System Level Control Registers (slcr) • TMU (temperature sensor) • XADC		

libethdrivers

Source	DornerWorks	Comment	Status	Task
/libs/sel4_util_libs /libethdrivers/CMakeLists.txt		Specific to the Xilinx Zynq-7000		• check the functionality of the drivers provided • check adaptation to Xilinx Zynq UltraScale+
	/libs/sel4_util_libs/libethdrivers/include /ethdrivers/zynqmp.h	Specific to the Xilinx Zynq UltraScale+ (provided by DornerWorks)		• check the functionality of the drivers provided • check adaptation to Data61 code
	/libs/sel4_util_libs/libethdrivers/plat_include /zynqmp/ethdrivers/plat/eth_plat.h	Specific to the Xilinx Zynq UltraScale+ (provided by DornerWorks)		• check the functionality of the drivers provided • check adaptation to Data61 code
	/libs/sel4_util_libs/libethdrivers/src/plat/zynqmp	Wrapper around the U-Boot Ethernet driver for the Xilinx Zynq UltraScale+		• check the functionality of the drivers provided • check adaptation to Data61 code

libsdhcd drivers

Currently, there is no support for SDHCI.

CAMKES

Source	Comment	Status	Task
/tools/camkes/libsel4camkes/src/sys_undef.c	only support for CONFIG_PLAT_ZYNQ7000		check if support is also required for the Zynq UltraScale+

Global Components

Component	Data61	DornerWorks	Comment	Status	Task
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Ethdriver		/libs/sel4_global_components /components/Ethdriver/include/plat /zynqmp/plat/eth_devices.h	specific support for Xilinx Zynq UltraScale+		• check the functionality of the driver provided • check adaptation on to Data61 code
		/libs/sel4_global_components /components/Ethdriver/src/plat/zynqmp /plat.c	specific support for Xilinx Zynq UltraScale+		• check the functionality of the driver provided • check adaptation on to Data61 code
SerialServer		/libs/sel4_global_components /components/SerialServer/CMakeLists. txt	specific support for the Avnet Ultra96 v2		check adaptation to Data61 code
		/libs/sel4_global_components /components/SerialServer/include/plat /zynqmp/plat/serial.h /libs/sel4_global_components /components/SerialServer/include/plat /ultra96v2/plat/serial.h	specific support for Xilinx Zynq UltraScale+, split into the SBCs Xilinx ZCU102 and Avnet Ultra96 v2		• check the functionality of the drivers provided • check adaptation on to Data61 code
		/libs/sel4_global_components /components/SerialServer/zynqmp /plat.c	specific support for Xilinx Zynq UltraScale+		• check the functionality of the driver provided • check adaptation on to Data61 code
TimeServer	/libs/sel4_global_components /components/TimeServer/include/plat /zynqmp/plat/timers.h		specific support for Xilinx Zynq UltraScale+		-
		/libs/sel4_global_components /components/TimeServer/src/plat /zynqmp/plat.c	specific support for Xilinx Zynq UltraScale+		• check the functionality of the drivers provided • check adaptation on to Data61 code

Device driver comparison

The following comparison table sums up the findings regarding the seL4 userspace libraries provided above and shows to which extent support already exists for the Xilinx Zynq UltraScale+ within several systems. As comparison, also the support for the predecessor Xilinx Zynq-7000 is provided, as Data61 supports a few more hardware components for this platform which might be a future target regarding porting.

HW Type	Zynq-7000 (Data61)	Zynq UltraScale+ (Data61)	Zynq UltraScale+ (Dornerworks)	U-Boot
Pinmux	/libs/seL4_util_libs/libplatsupport /mach_include/zynq/platsupport/mach/mux.h /libs/seL4_util_libs/libplatsupport/src/mach /zynq/mux.c	/libs/seL4_util_libs/libplatsupport /mach_include/zynq/platsupport/mach/mux.h /libs/seL4_util_libs/libplatsupport/src/mach /zynq/mux.c		
Timer	/libs/seL4_util_libs/libplatsupport /mach_include/zynq/platsupport/mach/timer.h /libs/seL4_util_libs/libplatsupport/plat_include /zynq7000/platsupport/plat/timer.h /libs/seL4_util_libs/libplatsupport/src/mach /zynq/timer.c /libs/seL4_util_libs/libplatsupport/src/mach /zynq/timer.c	/libs/seL4_util_libs/libplatsupport /mach_include/zynq/platsupport/mach/timer.h /libs/seL4_util_libs/libplatsupport/plat_include /zynqmp/platsupport/plat/timer.h /libs/seL4_util_libs/libplatsupport/src/mach /zynq/timer.c /libs/seL4_util_libs/libplatsupport/src/mach /zynq/timer.c		
Clock	/libs/seL4_util_libs/libplatsupport/plat_include /zynq7000/platsupport/plat/clock.h /libs/seL4_util_libs/libplatsupport/src/plat /zynq7000/clock.c	/libs/seL4_util_libs/libplatsupport/plat_include /zynqmp/platsupport/plat/clock.h /libs/seL4_util_libs/libplatsupport/src/plat /zynqmp/clock.c		
UART	/libs/seL4_util_libs/libplatsupport /mach_include/zynq/platsupport/mach /axi_uartlite.h /libs/seL4_util_libs/libplatsupport/plat_include /zynq7000/platsupport/plat/serial.h /libs/seL4_util_libs/libplatsupport/src/mach /zynq/axi_uartlite.c /libs/seL4_util_libs/libplatsupport/src/mach /zynq/chardev.c /libs/seL4_util_libs/libplatsupport/src/mach /zynq/serial.c	/libs/seL4_util_libs/libplatsupport /mach_include/zynq/platsupport/mach /axi_uartlite.h /libs/seL4_util_libs/libplatsupport/plat_include /zynqmp/platsupport/plat/serial.h /libs/seL4_util_libs/libplatsupport/src/mach /zynq/axi_uartlite.c /libs/seL4_util_libs/libplatsupport/src/mach /zynq/chardev.c /libs/seL4_util_libs/libplatsupport/src/mach /zynq/serial.c	/libs/seL4_util_libs/libplatsupport /plat_include/zynqmp/platsupport/plat /serial.h	drivers/serial /serial_zynq.c
Interrupt	ARM GIC	ARM GIC		
Reset	/libs/seL4_util_libs/libplatsupport/plat_include /zynq7000/platsupport/plat/src.h /libs/seL4_util_libs/libplatsupport/src/plat /zynq7000/src.c /libs/seL4_util_libs/libplatsupport/src/plat /zynq7000/src.h	/libs/seL4_util_libs/libplatsupport/plat_include /zynqmp/platsupport/plat/src.h		
Ethernet	/libs/seL4_util_libs/libethdrivers/include /ethdrivers/zynq7000.h /libs/seL4_util_libs/libethdrivers/plat_include /zynq7000/ethdrivers/plat/eth_plat.h /libs/seL4_util_libs/libethdrivers/src/plat /zynq7000		/libs/seL4_util_libs/libethdrivers/include /ethdrivers/zynqmp.h /libs/seL4_util_libs/libethdrivers /plat_include/zynqmp/ethdrivers/plat /eth_plat.h /libs/seL4_util_libs/libethdrivers/src/plat /zynqmp	/drivers/net /zynq_gem.c

SDHCI	-	-	-	drivers/mmc/zynq_sdhci.c
(Optional) QSPI Flash	-	-	-	drivers/spi/zynqmp_gspi.c
(Optional) eMMC	-	-	-	
(Optional) GPIO	/libs/sel4_util_libs/libplatsupport/mach/gpio.h	/libs/sel4_util_libs/libplatsupport/mach/gpio.h		drivers/gpio/zynq_gpio.c
(Optional) I2C	/libs/sel4_util_libs/libplatsupport/mach/i2c.h /libs/sel4_util_libs/libplatsupport/plat_include/zynq7000/platsupport/plat/i2c.h	/libs/sel4_util_libs/libplatsupport/mach/i2c.h /libs/sel4_util_libs/libplatsupport/plat_include/zynqmp/platsupport/plat/i2c.h		drivers/i2c/i2c-cdns.c
(Optional) SPI	-	-	-	drivers/spi/zynq_spi.c
(Optional) USB	-	-	-	drivers/usb/dwc3
Additional	/libs/sel4_util_libs/libplatsupport/plat_include/zynq7000/platsupport/plat/tmu.h /libs/sel4_util_libs/libplatsupport/src/plat/zynq7000/devcfg.c /libs/sel4_util_libs/libplatsupport/src/plat/zynq7000/devcfg.h /libs/sel4_util_libs/libplatsupport/src/plat/zynq7000/tmu.c /libs/sel4_util_libs/libplatsupport/src/plat/zynq7000/xadc.c /libs/sel4_util_libs/libplatsupport/src/plat/zynq7000/xadc.h	-	-	

U-Boot

- Xilinx: <https://github.com/Xilinx/u-boot-xlnx>
- Mainline: <https://github.com/u-boot/u-boot>

Porting Overview

According to [SEOS-2163](#) - Getting issue details... STATUS, the following HW parts are mandatory for porting:

- Bootloader
- Clock
- Interrupt
- Timer
- UART
- Ethernet
- SDHCI

Porting Proposal

The porting will take place in several consecutive steps. First of all, the process can be split regarding the two affected boards.

- As the Xilinx ZCU102 is already supported in a basic form by Data61, porting will start with this development board for having a common baseline. Thes ZCU102 is available both as physical HW platform and as virtual target in QEMU.
- Afterwards, the already ported drivers can be taken over to the low-cost Avnet Ultra96 platform. Here, we have to take into account that two versions of the SBC with a slightly different hardware setup exist. QEMU support isn't available.

The overall porting process will therefore look like the following:

1. Reuse the existent or acquire new board(s) supported officially by Data61
 - a. Xilinx ZCU102
 - b. Avnet Ultra96

2. Start with porting to the Xilinx ZCU102 development board (which is the Data61 reference platform for the Xilinx Zynq UltraScale+ MPSoC)
 - a. Foundation: Bootloader
 - i. Reuse the Xilinx U-Boot bootloader
 - ii. Reuse (and adapt if necessary) the official (or mainline) U-Boot bootloader
 - a. Basic port - Part I: Reuse and test existing drivers for
 - i. Clock
 - ii. UART
 - iii. Reset
 - iv. Timer
 - v. Interrupt
 - vi. Pinmux
 - b. Basic port - Part 2: Reuse, port or create new drivers for
 - i. Ethernet
 - ii. SDHCI
3. Switch to an up-to-date QEMU version that supports the Xilinx ZCU102 and repeat the steps mentioned above with a focus on the QEMU specific aspects of the **xlnx-zcu102** platform
4. Switch to the Avnet Ultra96 and repeat the steps mentioned above

Porting Workflow

Epics

Type	T	Key	Status	P	Summary	Resolution	Assignee
Authenticate to retrieve your issues							
No issues found							

Provide TRENTOS-M basic support for the Xilinx ZCU102

Type	T	Key	Status	P	Summary	Resolution	Assignee
Authenticate to retrieve your issues							
No issues found							

Provide TRENTOS-M Ethernet support for the Xilinx ZCU102

Type	T	Key	Status	P	Summary	Resolution	Assignee
Authenticate to retrieve your issues							
No issues found							

Provide TRENTOS-M SDHCI support for the Xilinx ZCU102

Type T Key Status P Summary Resolution Assignee

[Authenticate](#) to retrieve your issues

No issues found

Provide the TRENTOS-M IoT Demo for the Xilinx ZCU102

Type T Key Status P Summary Resolution Assignee

[Authenticate](#) to retrieve your issues

No issues found

Provide TRENTOS-M support for the Avnet Ultra96

Type T Key Status P Summary Resolution Assignee

[Authenticate](#) to retrieve your issues

No issues found

Porting Plan

As we have three target platforms in total and not all tickets depend on each other, we are able to parallelize the overall porting process to a certain extent. We basically start with adding support to the Xilinx ZCU102 QEMU platform, as the ZCU102 is the Data61 reference platform for the Xilinx Zynq UltraScale+ MPSoC. Based on the resulting adaptations we can afterwards improve this support in an incremental fashion, and afterwards start working on the implementation of additional support for the physical ZCU102 development board. As a last step, we provide support to the low cost board Avnet Ultra96. Nevertheless, in the end all platform adaptations have to be integrated into a final version.

Porting Stage	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Ticket																	
SEOS-2287 - Getting issue details... STATUS	x																
SEOS-2290 - Getting issue details... STATUS		x															
SEOS-2292 - Getting issue details... STATUS			x														

SEOS-2296 - Getting issue details... STATUS			x														
SEOS-2294 - Getting issue details... STATUS				x													
SEOS-2298 - Getting issue details... STATUS				x													
SEOS-2300 - Getting issue details... STATUS				x													
SEOS-2302 - Getting issue details... STATUS					x												
SEOS-2288 - Getting issue details... STATUS						x											
SEOS-2289 - Getting issue details... STATUS							x										
SEOS-2291 - Getting issue details... STATUS								x									
SEOS-2293 - Getting issue details... STATUS									x								
SEOS-2297 - Getting issue details... STATUS										x							
SEOS-2295 - Getting issue details... STATUS											x						
SEOS-2299 - Getting issue details... STATUS												x					
SEOS-2301 - Getting issue details... STATUS													x				
SEOS-2303 - Getting issue details... STATUS														x			
SEOS-2304 - Getting issue details... STATUS															x		
SEOS-2305 - Getting issue details... STATUS																x	
SEOS-2306 - Getting issue details... STATUS																	x
SEOS-2307 - Getting issue details... STATUS																	x
SEOS-2308 - Getting issue details... STATUS																	x
SEOS-2309 - Getting issue details... STATUS																	x
SEOS-2310 - Getting issue details... STATUS																	x

Story	I2C support (ZCU102 + QEMU, Ultra96)	TODO DOD: • Driver should run on the ... board. • Driver should be implemented as a component in TRENTOS-M.	
Story	SPI support (ZCU102 + QEMU, Ultra96)	TODO DOD: • Driver should run on the ... board. • Driver should be implemented as a component in TRENTOS-M.	
Story	TRNG/Crypto (ZCU102 + QEMU, Ultra96)	TODO DoD: • analyze ... • compare other implementations (Linux, wolfSSL, NXP reference) • create a plan and estimation (Wiki Page) for implementation, for 2 sets of functionalities ◦ only RNG ◦ RNG and cryptographic functions ◦ crypto performance tests ◦ ...	
Story	eMMC (ZCU102 + QEMU, Ultra96)	TODO • Check if (SD card related) driver(s) from ... porting can be re-used • Provide ... eMMC controller driver ◦ provide eMMC library or use code from seL4 platsupport repos ◦ driver just wraps the libs ◦ can eMMC be simply seen as a non-removeable SD-Card ◦ document differences between SD-Card and eMMC access in the storage subsystem and update the Wiki DoD • read datasheet of ... board • test storage interface running on board with light adjustment of sdhc • document any blockers you might experience	

Testing Plan

- all tests that have no real dependencies and can run on its own (e.g. test_crypto_api)
 - demo_hello_world
 - test_certparser
 - test_certserver
 - test_config_server
 - test_crypto_api
 - test_cryptoserver
 - test_entropysource
 - test_filesystem
 - test_keystore
- all tests that require UART support
 - test_uart
- all tests that require ChanMux support
 - test_chanmux
 - test_proxy_nvm
- all tests that require Timer
 - test_logserver
 - test_timeserver
- all tests that require Storage (RamDisk, ChanMux and platform dependent)
 - test_storage_interface
- all tests that require Networking (ChanMux and platform dependent)
 - test_network_api
 - test_tls_api
 - test_tlsserver